International Journal of Heat and Fluid Flow 82 (2020) 108552

Contents lists available at ScienceDirect

International Journal of Heat and Fluid Flow

journal homepage: www.elsevier.com/locate/ijhff

Consistent surface roughness extension for wall functions

Aviral Prakash*, Eric Laurendeau

Department of Mechanical Engineering, Polytechnique Montreal, 2900 Edouard Montpetit Blvd, Montreal, QC H3T 1J4, Canada

ARTICLE INFO

Keywords:
Heat transfer
Turbulence modeling
Wall function
Wall roughness

ABSTRACT

An approach to extend the near-wall treatment approach of RANS turbulence models using wall functions for rough surfaces is proposed. The formulation is consistent with the low-Reynolds number turbulence model roughness extension, leading to reduced near-wall grid spacing dependence and accurate predictions for a large range of k_s^+/y^+ values. The formulation is further extended to the temperature wall functions and heat transfer prediction is assessed. The drawbacks of using the Reynolds analogy assumption are highlighted and three thermal corrections to improve these drawbacks are also examined. In particular, the correction term by Aupoix (2015), which accounts for the effect of the wetted surface area, is incorporated in the wall function formulation. This correction term significantly improves predictions as compared to the other two thermal correction terms at the expense of adding an additional physical parameter.

1. Introduction

Rough surfaces appear in several applications, situations involving flows over gas turbine blades with combustion deposits, degraded surfaces during hypersonic re-entry, aircraft components with iced surfaces being just a few amongst the many scenarios. In all these applications, the presence of roughness increases the skin-friction drag and the heat transfer.

In industries, the effect of roughness is often modeled using Reynolds Averaged Navier-Stokes (RANS) equations based approaches. Most commonly, the approach involves the use of equivalent sand-grain roughness (ESGR) which overcomes a need for a fine mesh resolving each roughness element. The ESGR approach can be incorporated in both low-Reynolds number (low-Re) and high-Reynolds number (Wall functions) turbulence model formulations. The first involves the use of a fine mesh to resolve the viscous sub-layer, whereas the second involves a coarser near-wall mesh that relies on so-called wall functions to estimate the near-wall boundary layer properties. The advantages of using coarser meshes are reduced computational stiffness, which is often linked with increases in flow residual convergence rates as well as reduced computational memory requirements. The residual convergence rates of simulations involving roughness extension for the low-Reynolds turbulence model further decrease with increases in the roughness heights, which increases computational costs in many industrial applications. Multi-layer ice-shape prediction is one such application where multiple RANS computations at large ESGR height are required and use of wall functions can significantly reduce the computational cost (Chedevergne, 2018). In this application, the predicted ice-shape and the resulting aerodynamic performance loss are very much dependent on the selected ESGR height (Kontogiannis et al., 2018) and therefore, an accurate estimation of the heat transfer coefficient is important. Keeping these factors in mind, an investigation into the use of wall functions for the prediction of skin-friction drag and heat transfer over rough surfaces is necessary.

Several wall function approaches to account for surface roughness effect exist. Analytical wall functions (AWF) devised by Craft et al. (2002) was extended for roughness effects by Suga et al. (2006). This roughness extension was empirically determined. Apsley (2007) determined a more physical relation for the extension by performing expansion of the log-law form and ensuring higher-order consistency. It is well known that Reynolds analogy, which is a viable assumption for modeling heat transfer over smooth surfaces, does not hold in presence of rough surfaces (Cope, 1941). This argument is in coherence with the experimental results by Dipprey and Sabersky (1963) and Owen and Thomson (1963). To overcome this assumption for the low-Re turbulence model approach, Aupoix (2015) devised a correction term for turbulent Prandtl number based on numerical experiments using the Discrete Element Method (DEM). This correction introduces an additional parameter known as the corrected surface area ratio (S_{corr}) to improve heat transfer predictions. Even though the extension by Suga et al. (2006) overcomes this assumption by differential treatment of heat transfer, several issues persist with this formulation (Chedevergne, 2018). One of the key issues is that the formulation by Suga et al. depends on a single roughness scaling

* Corresponding author.
E-mail addresses: aviral2000@gmail.com, aviral.prakash@polymtl.ca (A. Prakash).

https://doi.org/10.1016/j.ijheatfluidflow.2020.108552
Received 11 July 2019; Received in revised form 21 December 2019; Accepted 17 January 2020
Available online 31 January 2020
0142-727X/ © 2020 Elsevier Inc. All rights reserved.

parameter which may not be adequate for accounting for roughness shape effects. Chedevergne (2018) remarked the importance of the corrected surface area ratio (wetted area) and incorporated this in the roughness extension for AWF.

Even though, the use of AWF for simulating flows over rough surfaces yields promising results, it still suffers from some issues. Popovac and Hanjalic (2007) highlighted that the complexity of the AWF formulation makes the implementation in general-purpose CFD code unfavorable. This is supported by the fact that Chedevergne (2018) had to incorporate several modifications to use AWF in an unstructured grid solver. Another common issue with most wall function approaches is the near-wall spacing dependence of the solution. For smooth walls, Kalitzin et al. (2005) showed that the use of a coarser grid increases numerical errors. The deviation is higher in the buffer layer due to the non-linear variation of turbulence variables. For the case of rough surfaces, the near-wall spacing dependence is even more severe. Most commercial CFD solvers restrict that minimum near-wall spacing (y^+) should be preferably greater than the ESGR height (k_s^+) and employ a limiter on the values of k_s^+ to improve robustness. Suga et al. (2006) also recommended a similar condition to ensure accurate heat transfer estimation. Eça and Hoekstra (2011) studied the near-wall spacing dependence of the roughness extension by Apsley (2007) and arrived at a similar conclusion. As these quantities are not known apriori, this poses several restrictions on the mesh generation process. In general, there is a lack of a detailed investigation on the near-wall spacing dependence of wall functions in the presence of roughness, let alone taking into account heat transfer.

The present paper improves over these drawbacks associated with the roughness extension for wall functions. A roughness extension that is consistent with the low-Re turbulence model extension is proposed to reduce the near-wall grid spacing dependence. The consistency is ensured by combining the formulation for low-Re turbulence model roughness extension by Aupoix (2007) with the appropriate turbulence model boundary condition proposed by Kalitzin et al. (2005). Since the work by Kalitzin et al. (2005) only addressed smooth surfaces, the paper also extends their boundary conditions to rough surfaces. The final approach gives accurate results for skin-friction drag and heat transfer over a large range of k_s^+/y^+ ratios. Lastly, we incorporate in our model the corrected surface area ratio proposed by Aupoix (2015) which is immensely important for improving the prediction of heat transfer over rough surfaces.

The applicability of proposed approach for wall function roughness extension is validated for Spalart-Allmaras (Spalart and Allmaras, 1992) and $k-\omega$ SST (Menter, 1994) models. The paper starts by highlighting the common non-dimensional variables and the smooth wall function approach. Then, the effect of roughness on velocity and temperature profiles is discussed. Finally, the approach used for the extension of wall functions for rough walls is proposed and its performance for several test cases is assessed.

2. Smooth wall functions

2.1. Non-dimensional variables

The flow, turbulence and roughness variables can be non-dimensionalized using appropriate scaling variables to generalize the behavior in the inner region of the boundary layer. These variables are defined as follows,

$$u^+ = \frac{u}{u_\tau},\ y^+ = \frac{yu_\tau}{\nu},\ T^+ = \frac{T}{T_\tau}$$
$$\tilde{\nu}^+ = \frac{\tilde{\nu}}{\nu},\ \nu_t^+ = \frac{\nu_t}{\nu},\ k^+ = \frac{k}{u_\tau^2},\ \omega^+ = \frac{\omega\nu}{u_\tau^2}$$

where u_τ $(=(\tau_w/\rho)^{1/2})$ and T_τ $(=q_w/\rho u_\tau c_p)$ are called friction velocity and friction temperature respectively.

2.1.1. Wall function formulation

The present work focuses on the use of log-law based wall functions. Even though there are several other wall function formulations such as AWF, log-law based wall functions are commonly employed in most CFD solvers due to the ease of implementation facilitating greater robustness over unstructured 3D meshes. These wall functions rely on semi-analytical near-wall velocity and temperature profiles. The applicability of the present wall function formulation is demonstrated using commonly used profiles: the velocity profile by Spalding (1961) and the temperature profile by Jayatillaka (1969). These profiles are given in Eq. (1) and Eq. (2) respectively, with Pr_f given in Eq. (3).

$$y^+ = u^+ + e^{-\kappa B}\left[e^{\kappa u^+} - 1 - \frac{(\kappa u^+)}{1!} - \frac{(\kappa u^+)^2}{2!} - \frac{(\kappa u^+)^3}{3!} - \frac{(\kappa u^+)^4}{4!} - \frac{(\kappa u^+)^5}{5!}\right] \tag{1}$$

$$T^+ = Pr(u^+ + Pr_f) \tag{2}$$

$$Pr_f = 0.24\left[\left(\frac{Pr}{Pr_t}\right)^{3/4} - 1\right]\left[1 + 0.28\exp\left(-0.007\frac{Pr_t}{Pr}\right)\right] \tag{3}$$

These velocity and temperature profiles for the smooth walls form the basis of the roughness extension discussed in this paper. The proposed approach applies to other velocity and temperature profiles also. To reduce the inconsistency, the wall functions must be accompanied by a compatible set of boundary conditions for turbulence models. For smooth walls, Kalitzin et al. (2005) highlighted the role of off-wall boundary conditions for maintaining the consistency and reducing the near-wall spacing dependence of wall functions. Off-wall boundary condition means that the turbulence model equations in the wall layer are not solved and analytical values of the turbulence variables are used instead. Similar to the approach by Kalitzin et al., an off-wall boundary condition is used for both Spalart-Allmaras and $k-\omega$ SST turbulence model. The near-wall behavior of turbulence models stipulates the turbulent viscosity to be related to the gradient of the velocity profile using Eq. (4). This expression is used to obtain the value of ν_t, which is essential to determining the value of turbulence variables.

$$\nu_t^+ = \frac{dy^+}{du^+} - 1 \tag{4}$$

The analytical solution of the Spalart-Allmaras turbulence model in the inner region of the boundary layer defines the value of the SA turbulence variable ($\tilde{\nu}$) as,

$$(\tilde{\nu}^+)^4 = \nu_t^+((\tilde{\nu}^+)^3 + C_{v1}^3) \tag{5}$$

For $k-\omega$ SST turbulence model, a single analytical expression quantifying the near-wall values of k and ω does not exist. Analytical relations are available for the viscous sublayer (Eq. (6)) and logarithmic layer(Eq. (7)).

$$k^+ = C_k\, y^{3.23},\quad \omega^+ = \frac{6}{\beta_1(y^+)^2} \tag{6}$$

$$k^+ = \frac{1}{(C_\mu)^{1/2}},\quad \omega^+ = \frac{1}{\kappa(C_\mu)^{1/2}y^+} \tag{7}$$

Several blending formulations exist to interpolate the values of ω from the corresponding values in the viscous sublayer and logarithmic layer. In the present study, the blending formulation proposed by Menter et al. (2003) is used for ω equation. For k equation, they used a zero-Neumann boundary condition. Kalitzin et al. (2005) remarked that this boundary condition for k is not suitable for the buffer layer and proposed the determination of k from ω and ν_t. Therefore, the boundary conditions by Kalitzin et al. (2005) are used and are given in Eq. (8).

$$k^+ = \nu_t^+\omega^+,\quad \omega^+ = ((\omega_{visc}^+)^2 + (\omega_{log}^+)^2)^{1/2} \tag{8}$$

These formulation and boundary conditions are used for smooth wall simulation cases in Section 4.

3. Rough wall functions

3.1. Roughness effects

Before moving to the proposed consistent rough wall function formulation, the effect of roughness on velocity and temperature profile is discussed in detail. Note that the definition of non-dimensional variables mentioned in Section 2 holds for rough walls also. In addition to those variables, a non-dimensional equivalent sand-grain roughness height is also introduced as shown in Eq. (9).

$$k_s^+ = \frac{k_s u_\tau}{\nu} \tag{9}$$

3.1.1. Effect on velocity profile

The present paper focuses on the 'k-type' roughness (Perry et al., 1969) which leads to an increase in the skin-friction drag and heat transfer that can be determined using the ESGR height. Nikuradse (1933) showed that roughness leads to a shift in the velocity profile as shown in Fig. 1.

The shift in the velocity profile is often quantified by a roughness function (Δu^+). The roughness function obtained from the experiments by Nikuradse (1933) is given in Eq. (12).

$$u^+ = \frac{1}{\kappa}\ln y^+ + C - \Delta u^+ \tag{10}$$

$$B = \begin{cases} 5.5 + \frac{1}{\kappa}\ln k_s^+ & 1 < k_s^+ \le 3.5 \\ 6.59 + 1.52\ln k_s^+ & 3.5 < k_s^+ \le 7 \\ 9.58 & 7 < k_s^+ \le 14 \\ 11.5 - 0.7\ln k_s^+ & 14 < k_s^+ \le 68 \\ 8.48 & 68 < k_s^+ \end{cases} \tag{11}$$

$$\Delta u^+ = \frac{1}{\kappa}\ln k_s^+ + C - B \tag{12}$$

The value of Von-Karman constant (κ) in these equations is 0.4, instead of the conventional value (= 0.41). Most of the turbulence models are based on $\kappa = 0.41$ and the use of a different value of κ results in inconsistency (Aupoix, 2014). To maintain consistency with the velocity profile used in the turbulence models, $\kappa = 0.41$ is taken in the present study. A roughness function was proposed by Grigson (1992) based on the experiments by Colebrook and White (1937). This is referred to as Colebrook-Grigson roughness function and is given in Eq. (13).

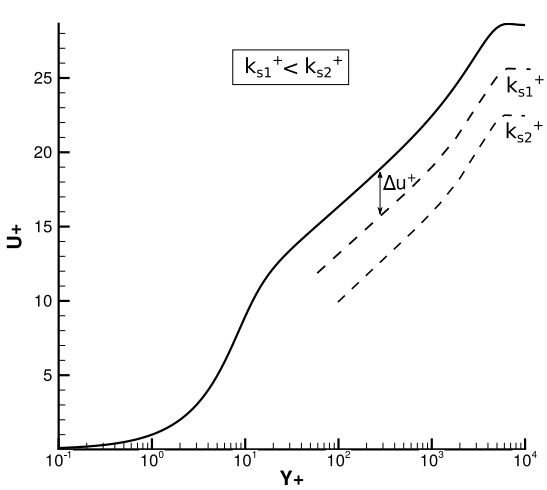


Fig. 1. Shift of the velocity profile due to roughness.

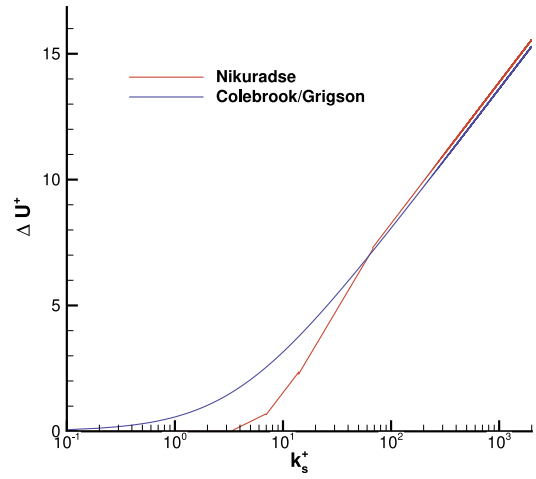


Fig. 2. Comparison of Nikuradse and Colebrook-Grigson roughness functions.

$$\Delta u^+ = \frac{1}{\kappa}\ln\left(1 + \frac{k_s^+}{e^{3.25\kappa}}\right) \tag{13}$$

The Nikuradse and Colebrook-Grigson roughness functions are compared in Fig. 2. The two roughness functions exhibit different behavior of friction drag in the transitionally rough regime, implying the effect of roughness element shape and distribution. The choice of roughness function in this regime depends on the roughness characteristics. For a general roughness pattern, a single roughness function is inadequate to give an accurate estimate in the transitionally roughness regime. Therefore, the present study focuses on the fully rough regime. The Colebrook-Grigson roughness function is used in the present study. However, the same approach can also be used with other roughness functions.

3.1.2. Effect on temperature profile

Roughness additionally results in the shift of the temperature profile. Even though the viscous sublayer is eliminated in the fully rough regime, the conduction sublayer still exists (Dipprey and Sabersky, 1963). Therefore, the maximum heat transfer through a rough surface is limited by conduction even when flow over roughness elements is separated (Owen and Thomson, 1963). The experimental study by Dipprey and Sabersky (1963) indicated that a certain combination of Reynolds number and Prandtl number leads to no further increase in Stanton number with the increase in k_s after a particular value of k_s, even though skin friction coefficient increases. All these observations indicate that the physics of momentum and heat transfer are not similar in the presence of roughness. This implies that the Reynolds analogy assumption, which is often used to model heat transfer for smooth surfaces, cannot be used in the presence of rough surfaces, as noted by Aupoix and Spalart (2003).

3.2. Proposed wall function approach

3.2.1. Velocity wall functions

The importance of the consistent wall function formulation for smooth walls was highlighted by Kalitzin et al. (2005) and Knopp et al. (2006). However, the present approach puts forward a slightly different take on it as it focuses on the consistency of the roughness extension. Kalitzin et al. (2005) showed that numerical errors arising from the use of a coarse grid reduce the accuracy of wall functions. As a result, even when the turbulence model exact velocity profiles (in form of a look-up table) are used for a smooth wall, the results do not coincide exactly with the low-Re turbulence model simulations for conventional wall functions grids. With the conventional

grids, the use of a slightly different velocity profile (i.e. velocity profile by Spalding) gives more accurate results. Therefore, this profile is used to demonstrate the extension of wall functions for rough surfaces. The approach to extend wall functions ensures consistency with the low-Reynolds number turbulence model roughness extension. The roughness extension for low-Reynolds turbulence model devised by Aupoix (2007) is based on the following relation,

$$\left.\frac{du_r^+}{dy^+}\right|_{y^+} = \left.\frac{du_s^+}{dy^+}\right|_{y^+ + y_0^+} \tag{14}$$

where y_0 is the virtual shift of the wall due to the presence of roughness (Rotta, 1962). Chedevergne and Aupoix (2017) integrated Eq. (14) and obtained the following form of velocity profile given in Eq. (15). This integration is based on the assumption of the presence of the fully rough regime and with the lower limit of velocity as $u_r^+ = 0$ at $y^+ = 0$.

$$u_r^+(y^+) = u_s^+(y^+ + y_0^+) - u_s^+(y_0^+) \tag{15}$$

$$\Delta u^+ = u_s^+(y_0^+) \tag{16}$$

Here, the subscripts r and s denote rough and smooth walls respectively. Eq. (15) gives the shifted velocity profile in the presence of roughness. In this study, Δu^+ is taken to be the Colebook-Grigson velocity shift (Eq. (13)). The corresponding smooth wall velocity profile (u_s^+) is evaluated using Eq. (1). Other semi-analytical velocity forms (i.e. $u_s^+(y^+)$) can also be used with this formulation. Using Eq. (16), the variation of y_0^+ with k_s^+ can be obtained for different combination of roughness functions and velocity profiles as shown in Fig. 3.

In Fig. 3, "Spalding-Cole" implies the use of Spalding velocity profile with Colebrook-Grigson roughness function. Similarly, the values obtained by using of Nikuradse roughness function with Spalding are denoted by "Spalding-Niku". These expressions are obtained by using Eq. (16) and the corresponding roughness functions. Lastly, "log-Niku" denotes the expression obtained by using Nikuradse roughness function with the log-law profile which is given as $y_0^+ = 0.03k_s^+$ (Aupoix and Spalart, 2003). The use of the virtual shift of the wall (y_0) is known as displacement of origin approach in some literature (Durbin, 2009). The use of this approach for wall functions has been used in the meteorological simulation community because they have empirically determined hydrodynamic roughness length (y_0) correlations (Richards and Hoxey, 1993). In the engineering community, equivalent sand-grain roughness (k_s) based approaches are mostly used instead of those using hydrodynamic roughness length. The present approach unifies these two roughness characteristic length scale by obtaining a relation between k_s^+ and y_0^+ based on standard roughness functions. The important aspect of using Eq. (16) is that it naturally reduces to low-Re turbulence model at low y+, thereby, reducing the near-wall grid spacing dependence.

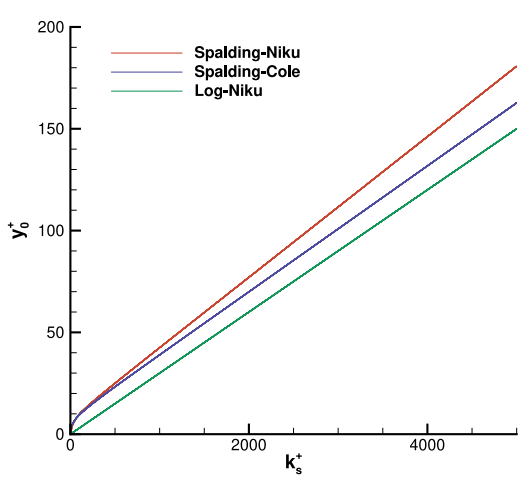
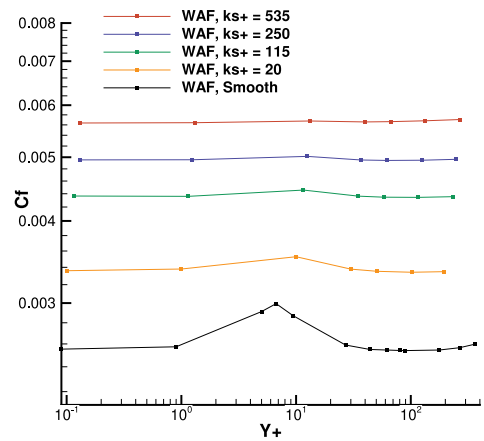


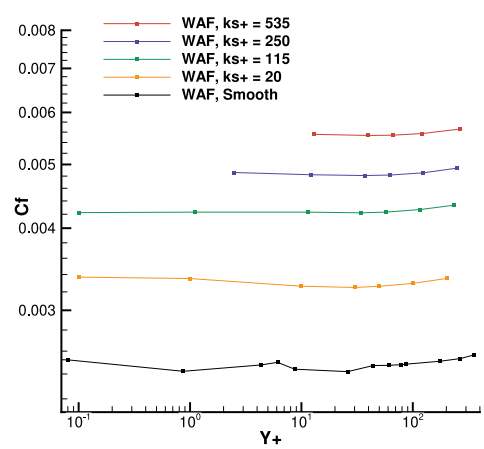
Fig. 3. Comparison of different analytic expressions of y_0^+.

Table 1
k_s^+ at $x = 1.5$ m for different values of k_s for the flat plate case.

k_s (m)	2 x 10^{-3}	1 x 10^{-3}	5 x 10^{-4}	1 x 10^{-4}
k_s^+	535	250	115	20



(a)



(b)

Fig. 4. Near-wall spacing dependence: C_f at x = 1.5 m for different values of y^+. (a) Spalart-Allmaras and (b) $k - \omega$ SST turbulence model.

3.2.2. Temperature wall functions

In the presence of roughness, the temperature profile also exhibits a similar shift and can be represented by the following equation,

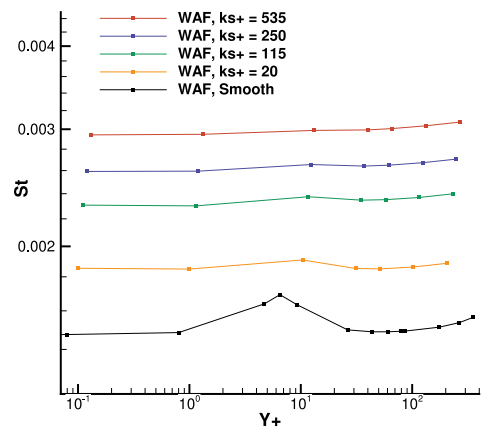
$$\left.\frac{dT_r^+}{dy^+}\right|_{y^+} = \left.\frac{dT_s^+}{dy^+}\right|_{y^+ + y_0^+} \tag{17}$$

Similar to the approach used for velocity profile, Eq. (17) can be integrated to obtain Eq. (18). Here, the lower limit of integration for T_r^+ is different i.e. at $y^+ = 0$, $T_r^+ = T_0^+$ due to the presence of a conduction sublayer.

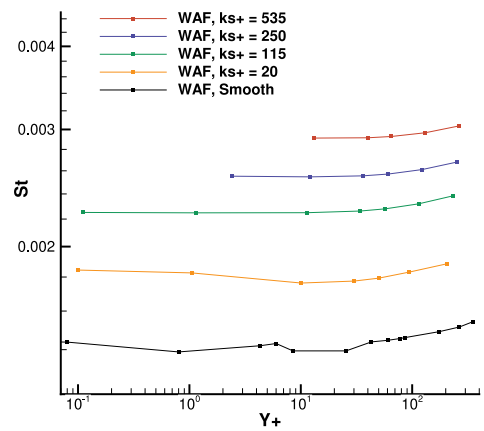
$$T_r^+(y^+) = T_s^+(y^+ + y_0^+) - \Delta T^+ \tag{18}$$

$$\Delta T^+ = T_s^+(y_0^+) - T_0^+ \tag{19}$$

Here, T_s is the smooth temperature profile shown in Eq. (2). The use of the assumption of Reynolds analogy can also be replicated by setting

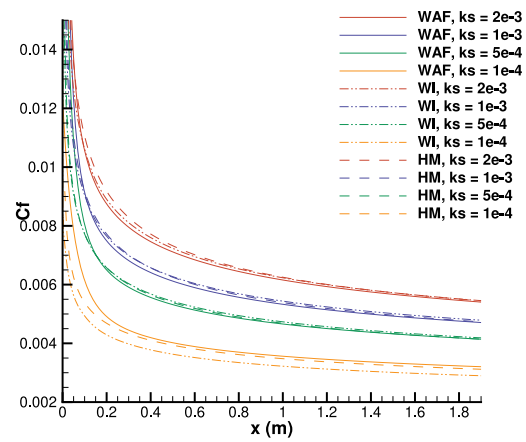


(a)

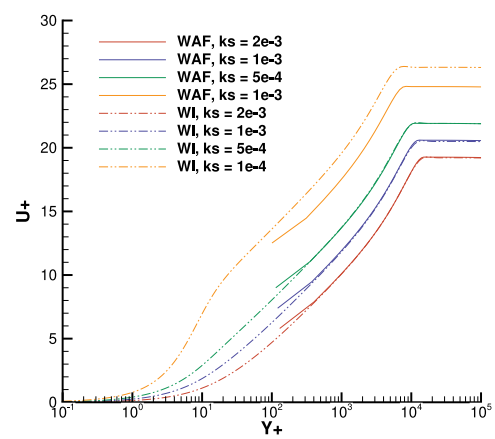


(b)

Fig. 5. Near-wall spacing dependence: St at x = 1.5 m for different values of y^+. (a) Spalart-Allmaras and (b) $k-\omega$ SST turbulence model.



(a)



(b)

Fig. 6. Mills-Hang case: (a) C_f variation and (b) velocity profile (at x = 1.9 m) for Spalart-Allmaras turbulence model.

$T_0^+ = 0$, implying that the conduction sublayer does not exist for the fully rough regime. However, as discussed earlier, this assumption is inadequate and results in the overestimation of St. Therefore, accurate models for T_0^+ are needed. Several empirical models exist for T_0^+, amongst which the earliest ones were proposed by Dipprey and Sabersky (1963) and Owen and Thomson (1963). These models can be expressed in the following form,

$$T_0^+ = \frac{1}{St_k} = \frac{1}{C_{th}} Re_k^m Pr^n \tag{20}$$

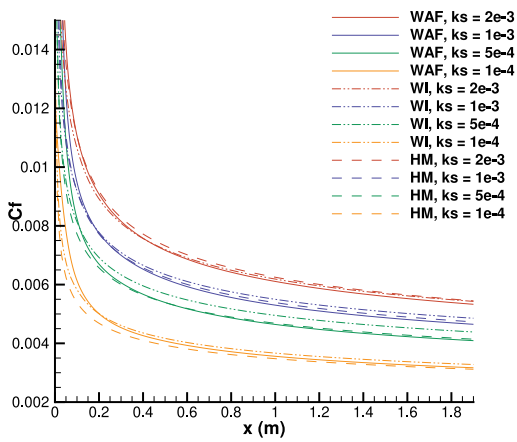
where, values of m and n are assigned as 0.2 and 0.44 respectively by Dipprey and Sabersky (1963). A commonly used value for C_{th} is 0.8 which is based on the experiments over spherical roughness elements (Pimenta et al., 1975). On the other hand, Owen and Thomson (1963) considered the 3-D flow across the cavities and determined the coefficients m and n to be 0.45 and 0.8 respectively. In this case, the value of C_{th} is taken as 1.92 for randomly shaped roughness. These two models are essential for accurate modeling of heat transfer and are thereby referred to as thermal correction models. Recently, Aupoix (2015) devised a new thermal correction model for overcoming the Reynolds analogy assumption for low-Re turbulence models. He used a database of Discrete Element Method (DEM) simulations and formulated a correction term to Pr_t in the boundary layer. He used additional physical parameters i.e. roughness height (h) and corrected surface area ratio (S_{corr}) to correlate the effect of roughness on heat transfer. Chedevergne (2018) showed that S_{corr} accounts for the effect of shape on heat transfer and improves heat transfer predictions. Radenac et al. (2018) extended the model by Aupoix to the Integral Boundary Layer (IBL) formulation. The thermal correction they devised is given as,

$$T_0^+ = \frac{F(k_s^+, S_{corr})}{\kappa} e^{y_0^+/h^+} \left[Ei\left(-\frac{y^+ + y_0^+}{h^+}\right) - Ei\left(-\frac{y_0^+}{h^+}\right) \right] \tag{21}$$

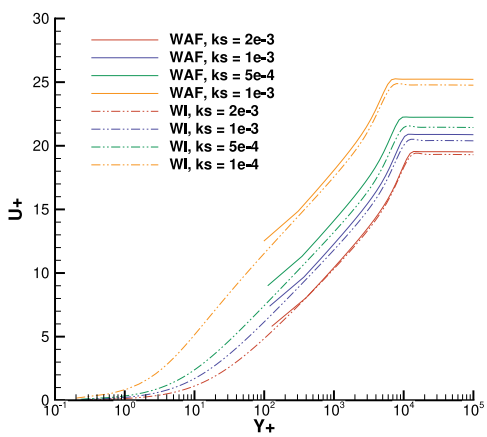
where, $h^+ = \frac{hu_\tau}{\nu}$ and F is the same function as used by Aupoix (2015) in his original model. Ei is the exponential integral function which, in the present paper, is obtained using tabulated data. This formulation of T_0^+ is exported to the proposed wall function approach. This improves the ability of wall functions to accurately predict heat transfer over rough surfaces. Note that the temperature profile used by Kays and Crawford (1993) and Radenac et al. (2018) can also be reduced to the form given in Eq. (18).

3.2.3. Turbulence model modifications

To ensure consistency with the low-Re turbulence model formulation, appropriate boundary conditions for turbulence model equations must be assigned. For smooth walls, the inconsistency between boundary condition for turbulence model equations increases near-wall spacing dependence of wall functions and leads to numerical errors (Durbin, 2009). The same reasoning applies to the treatment of

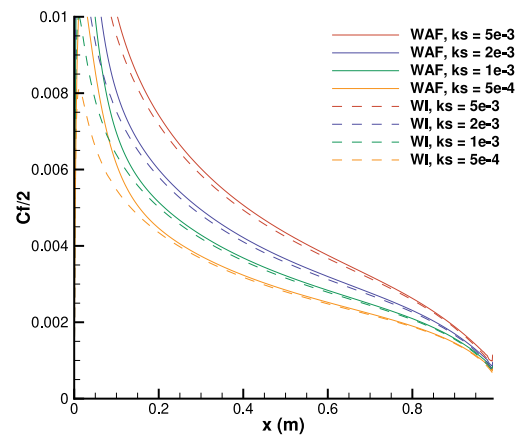


(a)

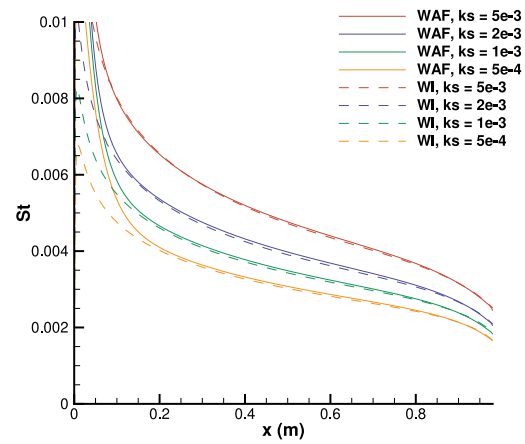


(b)

Fig. 7. Mills-Hang case: (a) C_f variation and (b) velocity profile (at x = 1.9 m) for $k-\omega$ SST turbulence model.



(a)



(b)

Fig. 8. Rough NACA 0012 airfoil case: (a) C_f and (b) St variation for Spalart-Allmaras turbulence model.

Table 2
k_s^+ at x = 0.5 m for different values of k_s for NACA 0012 airfoil case.

k_s (m)	5×10^{-3}	2×10^{-3}	1×10^{-3}	5×10^{-4}
k_s^+	1910	700	330	155

turbulence variables in most rough wall function formulations. This part of wall function formulation is often neglected or hidden in the literature and could be the reason for higher near-wall spacing dependence. Similar to the smooth wall formulation, off-wall boundary conditions are used for the turbulence variables. The eddy viscosity is defined similarly to the low-Re turbulence model roughness extension approach and is given by Eq. (22).

$$\nu_{t,r}^+(y^+) = \nu_{t,s}^+(y^+ + y_0^+) \tag{22}$$

For the Spalart-Allmaras turbulence model, the near-wall cell value for $\tilde{\nu}$ is given by Eq. (23). It was observed that imposing a Dirichlet boundary condition (Eq. (24)) at the wall for low values of y^+ gives a slightly better result. Similar to the low-Re turbulence model extension, the wall distance must be modified as shown in Eq. (25),

$$\tilde{\nu}_r^+(y^+) = \tilde{\nu}_s^+(y^+ + y_0^+) \tag{23}$$

$$\tilde{\nu}_{r,w}^+ = \tilde{\nu}_s^+(y_0^+) \tag{24}$$

$$d = y + y_0 \tag{25}$$

where the w subscript indicates the wall value. $\tilde{\nu}_s$ is obtained by solving Eq. (5). Unlike the BOEING low-Re turbulence model extension by Aupoix and Spalart (2003), the damping function should not be modified. The value of y_0 is determined from Eq. (16). This introduces a non-local parameter in the turbulence model computation. Even though the use of an analytical form of y_0 circumvents this additional computational cost, it results in inconsistency with the velocity profile leading to higher near-wall spacing dependence at low y^+.

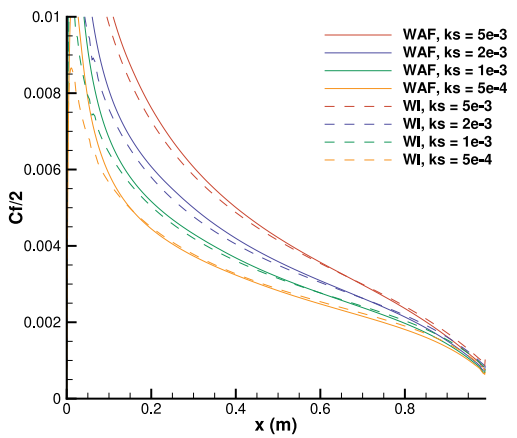
For $k-\omega$ SST turbulence model, the values of turbulence variables at the near-wall cell are defined as shown in Eq. (26). The modification of wall distance does not change the results significantly. The blending function for the buffer layer (Eq. (8)) is used to interpolate the value for the transitionally rough regime. The use of this interpolation function in the rough wall formulation improves the near-wall spacing dependence at low y^+ values for the transitionally rough regime.

$$k^+ = \omega^+\nu_t^+(y^+ + y_0^+), \quad \omega^+ = \frac{1}{\kappa(C_\mu)^{1/2}(y^+ + y_0^+)} \tag{26}$$

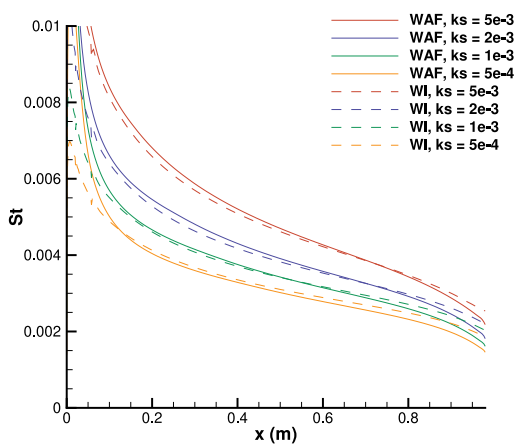
The same reasoning can be used for extending this wall function roughness extension in other turbulence models also.

4. Verification and validation

The simulations are performed using NSCODE which is an in-house

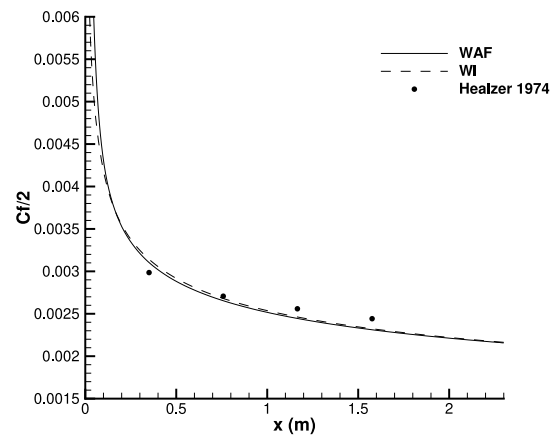


(a)

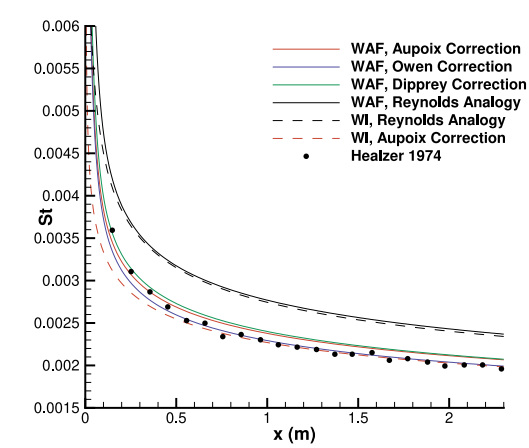


(b)

Fig. 9. Rough NACA 0012 airfoil case: (a) C_f and (b) St variation for $k-\omega$ SST turbulence model.



(a)



(b)

Fig. 10. Healzer et al. case: (a) C_f and (b) St variation along the flat plate using Spalart-Allmaras turbulence model.

structured multi-block 2/2.5D compressible RANS solver. More details about the solver can be found in Pigeon et al. (2014). The low-Reynolds number Spalart-Allmaras and $k-\omega$ SST turbulence models are extended using BOEING (Aupoix and Spalart, 2003) and "Colebrook" type (Aupoix, 2014) roughness extensions respectively. For the k equation in the low-Re $k-\omega$ SST turbulence model, a Dirichlet boundary condition is normally imposed for rough walls. However, convergence issues are observed at higher values of k_s. Therefore, the Neumann boundary condition is used instead to improve the numerical robustness (Fuhrman et al., 2010). The implementation of wall functions for a smooth surface is similar to the one mentioned in Carlson et al. (2015). For extending the implementation for rough walls, some changes are required.

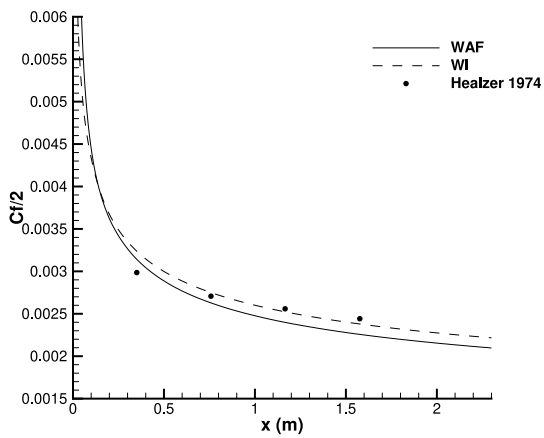
- For the combination of u_s^+ and Δu^+ chosen in this paper, y_0^+ can be obtained using Eq. (16). If other velocity profile such as the one by Reichardt (1951) is considered, then Eq. (16) needs to solved using Newton-Raphson to get the value of y_0^+.
- Eq. (15) is solved using Newton-Raphson to obtain the corresponding u_r^+ and y^+.
- Turbulence variables are updated using the consistent boundary conditions described in Section 3.2.3.
- T^+ is obtained from Eq. (18) using the y^+ obtained from step 2.
- Wall shear stress and wall heat flux are obtained from u_τ and T_τ respectively. These fluxes are added to the RANS equations.

The wall function roughness extension is verified with semi-empirical relation and low-Re turbulence model results in Sections 3.1-3.4 for the flat plate and NACA 0012 airfoil geometries. Further, the validation is provided against the experimental data in Sections 3.5 and 3.6. More validation cases for the Spalart-Allmaras model are shown in Prakash and Laurendeau (2019). In the figures, WI denotes the low-Re turbulence model simulation results and WAF indicates the wall function results. The flat-plate domain is similar to the one in NASA Turbulence Modeling Resource (TMR) (Center, 2018), however, with the far-field location at a height of 20m. Inflow and outflow are used as the inlet and outlet boundary conditions respectively. There are 257 points in both streamwise and wall-normal directions. For the NACA 0012 airfoil, an O-grid with around 700 points in streamwise and 129 in wall-normal directions are used. For all the simulations, a residual convergence of 10^{-6} is ensured for the density and turbulence variables.

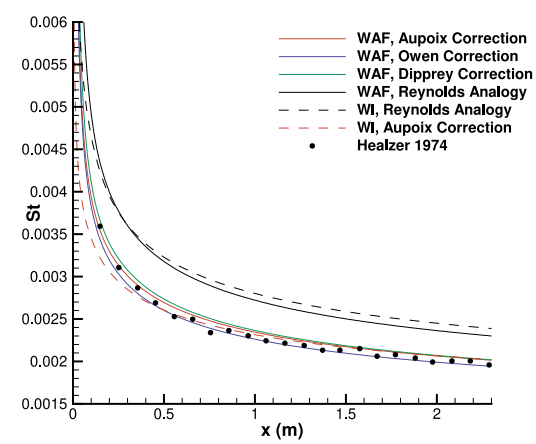
4.1. Near-wall spacing dependence

Analysis of the near-wall spacing dependence of wall functions is carried out by comparing the flat-plate results with meshes consisting of different y^+ at the first cell center. The case parameters are the same as the smooth zero 2-D ZPG (zero-pressure gradient) flat plate case in NASA TMR. Several values of k_s covering both the transitionally and the fully rough regimes are tested. These values are mentioned in Table 1.

The results for the Spalart-Allmaras turbulence model are shown in

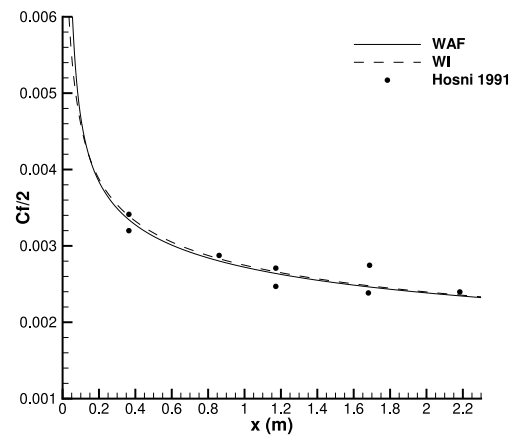


(a)

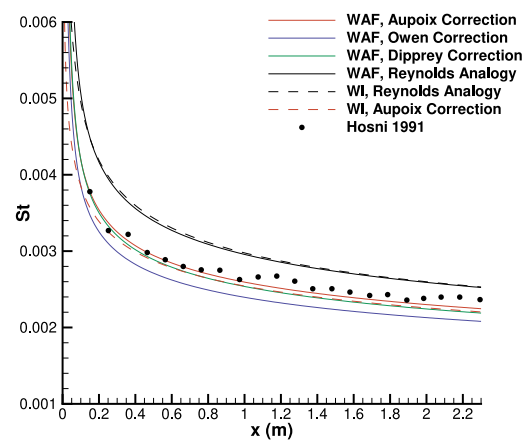


(b)

Fig. 11. Healzer et al. case: (a) C_f and (b) St variation along the flat plate using $k-\omega$ SST turbulence model.



(a)



(b)

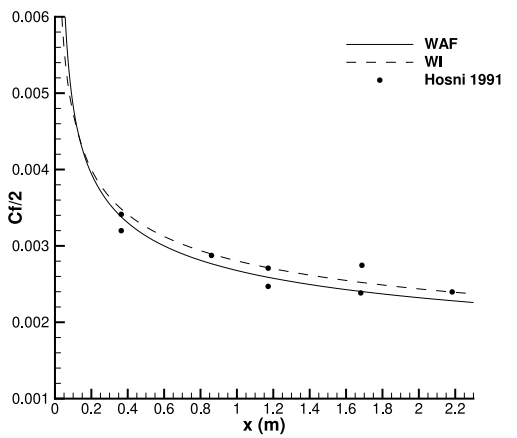
Fig. 12. Hosni et al. case: (a) C_f and (b) St variation for hemispherical elements with spacing of $l/d = 2$ using Spalart-Allmaras turbulence model.

Table 3
Roughness parameters for experiments by Hosni et al. (1993).

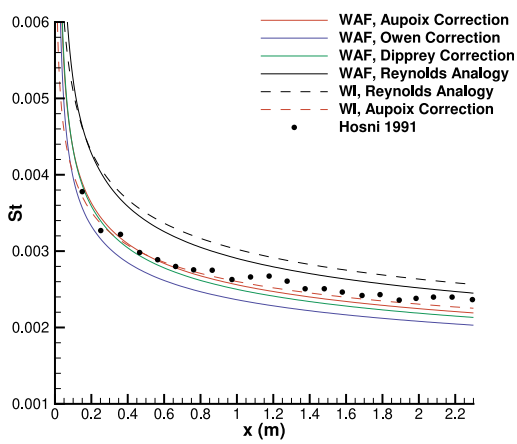
Element Type	k_s (mm)	h (mm)	S_{corr}
Cone	1.16	0.635	1.09
Hemisphere	1.16	0.635	1.17

Fig. 4(a). For smooth walls, a large deviation in the result is observed (represented by a peak) when the first cell center lies in the buffer layer ($5 < y^+ < 30$). A similar deviation, albeit shown for the velocity profile, was also observed by Kalitzin et al. (2005). This deviation is due to numerical errors resulting from the use of coarse grids. With the increase in the value of k_s, the size of the peak decreases and it is eliminated at higher values of k_s (in fully rough regime). Even at a high value of ESGR height ($k_s^+ = 535$), the results corresponding to meshes with a near-wall spacing of $y^+ \approx 0.1$ and $y^+ \approx 200$ are very similar. In most literature of rough wall functions, the use of a mesh with y^+ greater than the k_s^+ i.e. $k_s^+/y^+ < 1$ is recommended. Similar restrictions were mentioned by Eça and Hoekstra (2011) and Suga et al. (2006). The present wall functions approach overcomes this limitation and significantly increase the range (k_s^+/y^+) of applicability. As the viscous sublayer is eliminated in the fully rough regime, the logarithmic region extends to the wall and thus the deviation in results in the buffer layer is also eliminated. This gives a physical justification for the behavior of the roughness extension. The results for $k-\omega$ SST turbulence model are shown in Figure 4(b). In this case, the deviation in the buffer layer is not as significant as in the case of the Spalart-Allmaras turbulence model. The results show an insignificant deviation for a large range of near-wall spacing at different values of k_s^+. However, in this case, numerical convergence deterioration is observed with the current flow solver when a low y^+ is used at a large k_s^+. Still, the roughness extension is applicable for a larger range of k_s^+/y^+ as compared to other formulations in literature.

The variation of St for different values of near-wall spacing at various roughness heights is shown in Fig. 5. These results are obtained using the Reynolds analogy assumption. The behavior is similar for the other thermal corrections also. The near-wall spacing dependence for St is similar to the one obtained for C_f distribution. Slightly higher deviations are observed which is dependent upon the selection of the underlying smooth temperature profile. This is justified by a similar deviation behavior for both smooth and rough walls. Nevertheless, the results are quite accurate for a large range of k_s^+/y^+. The reduced near-wall spacing dependence increases the capability of the proposed wall functions to accurately predict the friction drag and heat transfer over complex geometries, in which maintaining close control over the y^+ values is not practical. Based on this analysis, meshes with $y^+ \approx 120$ are used for the validation cases.

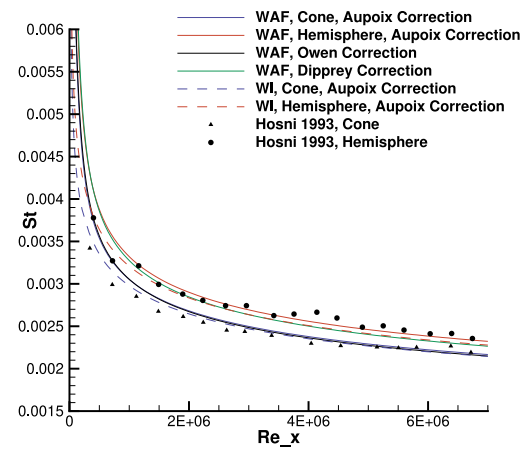


(a)

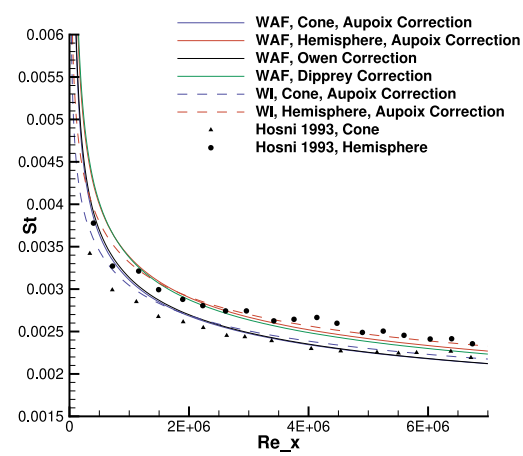


(b)

Fig. 13. Hosni et al. case: (a) C_f and (b) St variation for hemispherical elements with spacing of $l/d = 2$ using $k - \omega$ SST turbulence model.



(a)



(b)

Fig. 14. Hosni et al. case: (a) Spalart-Allmaras and (b) $k - \omega$ SST turbulence models results comparing the effect of roughness shape on St.

4.2. Semi-empirical relation by Mills and Hang (1983)

Mills and Hang (1983) combined the velocity profile in the inner region with the wake profile and numerically integrated it to obtain the variation of the skin-friction coefficient on the surface of flat-plate for different values of k_s. The relation is shown in Eq. (27) and is based on the assumption of the fully rough regime. The velocity profile used by them accounted for roughness using the Nikuradse roughness function (Nikuradse, 1933). It is relevant to note that they used $\kappa = 0.41$ for their roughness extension. Wall function results with Colebrook-Grigson roughness function are compared with the semi-empirical relation. Although the use of Nikuradse roughness function with $\kappa = 0.41$ is better suited for comparison and gives better agreement, the difference between the two roughness functions is small in the fully rough regime. The values of the corresponding k_s^+ are shown in Table 1. The flow conditions for this case are the same as the near-wall spacing dependence study. The semi-empirical relation results are denoted by "HM" in the figures.

$$C_f = \left[3.476 + 0.707\ln\left(\frac{x}{k_s}\right)\right]^{-2.46} \tag{27}$$

The results for Spalart-Allmaras and $k - \omega$ SST turbulence models are shown in Fig. 6 and Fig. 7 respectively. For the Spalart-Allmaras turbulence model, it is observed that the skin-friction coefficient and the velocity profile for wall function simulations match well with both the low-Re turbulence model simulations and the semi-empirical relation for the fully rough regime. Note that the semi-empirical relation by Mills and Hang (1983) is only valid for the fully rough regime and the transitionally roughness regime results are added only for completeness. In the transitionally rough regime, the difference in results between the wall function and low-Reynolds number turbulence model simulations is expected. The present roughness extension approach is valid for the fully rough regime and its use in the transitionally rough regime is questionable. Good agreement between the wall function results and the semi-empirical relation is also observed for $k - \omega$ SST turbulence model for all the roughness heights. These results match well with the low-Re turbulence model simulations at higher values of k_s^+. At lower values, there is a slight deviation between wall function and low-Re turbulence model approaches. More detailed analysis shows that this difference can be attributed to the use of a Neumann boundary condition instead of the traditional Dirichlet boundary condition for the k equation in the low-Re turbulence model roughness extension.

4.3. Low-Reynolds number simulation for Rough NACA 0012

Experimental data containing combined skin-friction coefficient and Stanton number distributions over rough airfoil surface is lacking in the literature. Most of the studies have focused on the aggregate coefficient of drag (C_d) to assess the effect of roughness on drag performance. Due

to the lack of experimental data, the low-Reynolds number turbulence model simulations are used for validating the roughness extension of the wall function approach. Here, the Reynolds analogy assumption is used to compare heat transfer results. The test case conditions are the same as the 0° Angle of Attack (AoA) case for NACA 0012 mentioned in NASA TMR. Various ESGR heights and corresponding values of k_s^+ are shown in Table 2.

The results for Spalart-Allmaras and $k-\omega$ SST turbulence models are shown in Fig. 8 and Fig. 9 respectively. For both turbulence models, the skin-friction coefficient and the Stanton number results for the wall function approach match well with the low-Re turbulence model simulations except at the leading edge region. This behavior is similar for all the tested values of roughness. At the leading edge i.e. low local Reynolds number region, the log-law does not hold. The wall function approach assumes fully turbulent flow (existence of log-law) even at the low local Reynolds number and therefore, it predicts higher skin-friction coefficient and Stanton number distributions. Even the roughness extension for low-Re turbulence models are based on the shift of the velocity profile and assumptions behind these extensions is the existence of the log-law region (Aupoix, 2007). Therefore, the use of these roughness extensions for flows involving significant adverse pressure gradients or a low Reynolds number, i.e. in scenarios where the log-law does not hold, is questionable. Considering this aspect, it can be said that within the assumption space of the wall function approach, the results are accurate.

4.4. Experiments by Healzer et al. (1974)

Healzer et al. (1974) conducted experiments on flows over rough surfaces at several flow velocities. In this paper, a velocity of 73.8m/s is considered. The surface was covered with closely packed spherical roughness elements of 0.635mm diameter. The equivalent sand-grain roughness was taken to be 1.25 times the diameter of the roughness element (Aupoix, 2015). As measurements of both C_f and St were given, this case is well suited for validating the thermal correction by Aupoix (2015).

From Fig. 10, it is observed that the Spalart-Allmaras turbulence model gives a good prediction for C_f. However, the use of the Reynolds analogy overestimates St. The use of different thermal corrections significantly improves the result and leads to closer predictions compared to experimental results. Low-Re turbulence model and wall function results are close to each other for both the Reynolds analogy and Aupoix thermal correction. For $k-\omega$ SST turbulence model, the results are similar and presented in Fig. 11.

4.5. Experiments by Hosni et al. (1993)

Hosni et al. (1991) conducted experiments over a flat-plate covered with hemispherical roughness elements at different spacing ratios. Measurement of both the skin-friction coefficient and Stanton number distributions were taken at several flow velocities. The study was later extended to include conical roughness elements and account for the effect of roughness element shape on heat transfer. The conical element configuration yielded similar skin-friction coefficient distribution. However, Hosni et al. (1993) observed that the Stanton number distribution for the two different types of roughness element shapes was not similar. In the present paper, the case of spacing ratio with $l/d = 2$ at a velocity of 58 m/s is considered. Similar validation was also performed by Chedevergne (2018) for his wall function approach. He provided the roughness parameters for both the roughness element shapes. Unlike the simulations by Chedevergne, the values of k_s are taken to be the experimental value (also used in the roughness extension study by Aupoix (2014)) to validate the results. The roughness parameters are highlighted in Table 3. The value of k_s^+ for this case is around 200.

The results for hemispherical roughness elements are shown in Fig. 12 and Fig. 13 for Spalart-Allmaras and $k-\omega$ SST turbulence models respectively. The results are similar for both turbulence models. The skin-friction coefficient prediction by the wall function approach is close to the low-Re turbulence model simulations and the experimental results. The use of the Reynolds analogy assumption leads to the overprediction of Stanton number distributions. This is also evident from other studies concerning heat transfer over rough surfaces (Aupoix and Spalart, 2003). The use of thermal corrections significantly improves the Stanton number prediction. The Stanton number predictions using Aupoix or Dipprey thermal corrections are close to the experimental results. The use of Owen thermal correction slightly underestimates the Stanton number. For $k-\omega$ SST turbulence model, the C_f and St are also in good agreement. For $k-\omega$ SST turbulence model, the prediction matches more closely with those from low-Re turbulence model simulations if the Nikuradse roughness function is used. However, these results are not shown in the present paper for brevity.

A comparison of the experimental Stanton number distributions and the RANS predictions for the two different roughness element types is shown in Fig. 14. It is observed that for both turbulence models, Aupoix thermal correction provides the best agreement. The effect of the wetted surface area included using the corrected surface area ratio parameter, significantly improves the results and accurately predicts the Stanton number for both roughness element types. The wall function predictions for Aupoix thermal correction also matches well with the corresponding low-Re turbulence model results. On the other hand, the two other thermal corrections are unable to replicate this behavior. The Dipprey correction results correspond to the experimental results for hemispherical roughness elements, whereas the Owen correction leads to predictions closer to the results for conical elements. This indicates that the use of a single roughness parameter (i.e. k_s) is inadequate for accurately predicting heat transfer over rough surfaces.

5. Conclusions

In this paper, wall functions are extended to incorporate the effect of surface roughness. The applicability of the formulation is shown for the Spalart-Allmaras and $k-\omega$ SST turbulence models. The proposed methodology blends the low-Re turbulence model extension of Aupoix (2007) with the turbulence model boundary conditions of (Kalitzin et al., 2005). This approach ensures consistency with the low-Re turbulence model roughness extension. This consistent formulation decreases the near-wall spacing dependence and allows the use of a larger range of k_s^+/y^+ ratios compared to other wall function formulations in the literature. The absence of the buffer layer in the fully rough regime is shown to reduce the near-wall spacing dependence compared to smooth walls. This characteristic is desirable when simulating fully turbulent attached flows over complex geometries, where the fine mesh resolution ($y^+ < 1$) is difficult to achieve. The skin-friction drag predictions obtained using the roughness correction are in good agreement with the low-Re turbulence model results, semi-empirical relation and experimental results.

The roughness extension is used in conjunction with the law of the wall for the temperature to predict heat transfer over rough surfaces. Reynolds analogy assumption is used to validate the roughness extension of wall functions with the low-Re turbulence model results. As the Reynolds analogy assumption does not hold in the presence of rough surfaces, three thermal corrections are investigated to improve the predictions of heat transfer. The thermal correction by Aupoix involves an additional physical parameter i.e. corrected surface area ratio for incorporating wetted surface area effects and gives better predictions as compared to the other two thermal corrections. The other thermal extensions give qualitatively good results, however, unlike the Aupoix correction they are unable to account for the effect of the roughness element shape on heat transfer.

The present wall function approach inherits some of the limitations of the underlying log-law based wall functions for smooth surfaces. The

AWF offers several advantages over conventional log-law based wall functions like improved predictions in the presence of pressure gradient and handling of buoyancy effects. However, AWF roughness extensions pose restriction on the selection of k_s^+/y^+ values. A comparative study between these two roughness extensions can shed more light on the advantages and disadvantages of these two approaches in the presence of roughness. The wall function formulation by Popovac and Hanjalic (2007), which exhibits good behavior is the presence of pressure gradients, resemble the log-law based wall functions. This wall function can be extended using a similar approach and the near-wall spacing dependence could be investigated. Moreover, the applicability of the proposed approach to other turbulence models can also be investigated.

CRediT authorship contribution statement

Aviral Prakash: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Eric Laurendeau:** Formal analysis, Investigation, Resources, Data curation, Writing - review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was sponsored by an NSERC Collaborative Research and Development Grant with Bombardier Aerospace under grant number CRDPJ-488646-2015. The first author also benefited from a Mitacs Globalink Graduate Fellowship award.

References

Apsley, D., 2007. Cfd calculation of turbulent flow with arbitrary wall roughness. Flow Turbul. Combust. 78 (2), 153–175. https://doi.org/10.1007/s10494-006-9059-x.
Aupoix, B., 2007. A general strategy to extend turbulence models to rough surfaces: application to Smith's k-L model. J. Fluids Eng. 129 (10), 1245. https://doi.org/10.1115/1.2776960.
Aupoix, B., 2014. Roughness corrections for the kω shear stress transport model: status and proposals. J. Fluids Eng. 137 (2), 21202. https://doi.org/10.1115/1.4028122.
Aupoix, B., 2015. Improved heat transfer predictions on rough surfaces. Int. J. Heat Fluid Flow 56, 160–171. https://doi.org/10.1016/j.ijheatfluidflow.2015.07.007.
Aupoix, B., Spalart, P.R., 2003. Extensions of the Spalart - Allmaras turbulence model to account for wall roughness. Int. J. Heat Fluid Flow 24 (4), 454–462. https://doi.org/10.1016/S0142-727X(03)00043-2.
Carlson, J.-R., Vatsa, V., White, J.A., 2015. Node-centered wall function models for the unstructured flow code Fun3D. 22nd AIAA Computational Fluid Dynamics Conference. American Institute of Aeronautics and Astronauticshttps://doi.org/10.2514/6.2015-2758. AIAA AVIATION Forum
Center, N.L.R., 2018. Turbulence Modelling Resource.
Chedevergne, F., 2018. Analytical wall function including roughness corrections. Int. J. Heat Fluid Flow 73 (August), 258–269. https://doi.org/10.1016/j.ijheatfluidflow.2018.08.001.
Chedevergne, F., Aupoix, B., 2017. Accounting for wall roughness effects in turbulence models : a wall function approach. 7th European Conference for Aeronautics and Aerospace Sciences (EUCASS). https://doi.org/10.13009/EUCASS2017-372.
Colebrook, C.F., White, C.M., 1937. Experiments with fluid friction in roughened pipes. Proc. R. Soc. Lond. A 161 (906), 367–381. https://doi.org/10.1098/rspa.1937.0150.
Cope, W.F., 1941. The friction and heat transmission coefficients of rough pipes. Proc. Inst. Mech.Eng. 145 (1), 99–105. https://doi.org/10.1243/PIME_PROC_1941_145_026_02.
Craft, T.J., Gerasimov, A.V., Iacovides, H., Launder, B.E., 2002. Progress in the generalization of wall-function treatments. Int. J. Heat Fluid Flow 23 (2), 148–160. https://doi.org/10.1016/S0142-727X(01)00143-6.
Dipprey, D., Sabersky, R., 1963. Heat and momentum transfer in smooth and rough tubes at various Prandtl numbers. Int. J. Heat Mass Transf. 6 (5), 329–353. https://doi.org/10.1016/0017-9310(63)90097-8.
Durbin, P.A., 2009. Limiters and wall treatments in applied turbulence modeling. Fluid Dyn. Res. 41 (1), 12203. https://doi.org/10.1088/0169-5983/41/1/012203.
Eça, L., Hoekstra, M., 2011. Numerical aspects of including wall roughness effects in the SST k-ω eddy-viscosity turbulence model. Comput. Fluids 40 (1), 299–314. https://doi.org/10.1016/j.compfluid.2010.09.035.
Fuhrman, D.R., Dixen, M., Jacobsen, N.G., 2010. Physically-consistent wall boundary conditions for the k-ω turbulence model. J. Hydraul. Res. 48 (6), 793–800. https://doi.org/10.1080/00221686.2010.531100.
Grigson, C., 1992. Drag losses of new ships caused by hull finish. J. Ship Res. 36, 182–196.
Healzer, J., Moffat, R., Kays, W., 1974. The turbulent boundary layer on a porous, rough plate - Experimental heat transfer with uniform blowing. Thermophysics and Heat Transfer Conference. American Institute of Aeronautics and Astronauticshttps://doi.org/10.2514/6.1974-680. Fluid Dynamics and Co-located Conferences
Hosni, M., Coleman, H., Taylor, R., 1991. Measurements and calculations of rough-wall heat transfer in the turbulent boundary layer. J. Heat Mass Transf. 34 (4–5), 1067–1082. https://doi.org/10.1016/0017-9310(91)90017-9.
Hosni, M.H., Coleman, H.W., Garner, J.W., Taylor, R.P., 1993. Roughness element shape effects on heat transfer and skin friction in rough-wall turbulent boundary layers. Int. J. Heat Mass Transf. 36 (1), 147–153. https://doi.org/10.1016/0017-9310(93)80074-5.
Jayatillaka, C., 1969. The influence of Prandtl number and surface roughness on the resistance of the laminar sub-layer to momentum and heat transfer. Progress Heat Mass Transf. 1:193.
Kalitzin, G., Medic, G., Iaccarino, G., Durbin, P., 2005. Near-wall behavior of RANS turbulence models and implications for wall functions. J. Comput. Phys. 204 (1), 265–291. https://doi.org/10.1016/j.jcp.2004.10.018.
Kays, W., Crawford, M., 1993. Convective Heat and Mass Transfer, third ed. McGraw-Hill, New-York.
Knopp, T., Alrutz, T., Schwamborn, D., 2006. A grid and flow adaptive wall-function method for RANS turbulence modelling. J. Comput. Phys. 220 (1), 19–40. https://doi.org/10.1016/j.jcp.2006.05.003.
Kontogiannis, A., Prakash, A., Laurendeau, E., Moens, F., 2018. Sensitivity of glaze ice accretion and iced aerodynamics prediction to roughness. 26th Annual Conference of the Computational Fluid Dynamics Society of Canada. Winnipeg, Manitoba.
Menter, F.R., 1994. Two-equation eddy-viscosity turbulence models for engineering applications. AIAA J. 32 (8), 1598–1605. https://doi.org/10.2514/3.12149.
Menter, F.R., Ferreira, J., Esch, T., Konno, B., 2003. The SST turbulence model with improved wall treatment for heat transfer predictions in gas turbines. International Gas Turbine Congress 2003. pp. 1–7.
Mills, A.F., Hang, X., 1983. On the skin friction coefficient for a fully rough flat plate. J. Fluids Eng. 105 (3), 364–365. https://doi.org/10.1115/1.3241008.
Nikuradse, J., 1933. Laws of flow in rough pipes. VDI Forschungsheft 361.
Owen, P.R., Thomson, W.R., 1963. Heat transfer across rough surfaces. J. Fluid Mech. 15 (3), 321–334. https://doi.org/10.1017/S0022112063000288.
Perry, A.E., Schofield, W.H., Joubert, P.N., 1969. Rough wall turbulent boundary layers. J. Fluid Mech. 37 (2), 383. https://doi.org/10.1017/S0022112069000619.
Pigeon, A., Levesque, A.T., Laurendeau, E., 2014. Two-dimensional Navier-Stokes flow solver developments at École polytechnique de montréal. CFD Society of Canada 22nd Annual Conference. Toronto, CA.
Pimenta, M.M., Moffat, R.J., Kays, W.M., 1975. The Turbulent Boundary Layer: An Experimental Study of the Transport of Momentum and Heat with the Effect of Roughness. Technical Report. Thermosciences Division, Department of Mechanical Engineering, Stanford University. HMT-21.
Popovac, M., Hanjalic, K., 2007. Compound wall treatment for RANS computation of complex turbulent flows and heat transfer. Flow Turbul. Combust. 78 (2), 177. https://doi.org/10.1007/s10494-006-9067-x.
Prakash, A., Laurendeau, E., 2019. A wall function approach to account for rough surface heat transfer in Spalart-Allmaras turbulence model. AIAA Aviation 2019 Forum. https://doi.org/10.2514/6.2019-2918.
Radenac, E., Kontogiannis, A., Bayeux, C., Villedieu, P., 2018. An extended rough-wall model for an integral boundary layer model intended for ice accretion calculations. Atmospheric and Space Environments Conference. https://doi.org/10.2514/6.2018-2858.
Reichardt, H., 1951. Vollstndige darstellung der turbulenten geschwindigkeitsverteilung in glatten leitungen. ZAMM 31 (7), 208–219. https://doi.org/10.1002/zamm.19510310704.
Richards, P., Hoxey, R., 1993. Appropriate boundary conditions for computational wind engineering models using the k-turbulence model. J. Wind Eng. Ind.Aerodyn. 46–47, 145–153. https://doi.org/10.1016/0167-6105(93)90124-7.
Rotta, J.C., 1962. Turbulent boundary layers in incompressible flow. Progress Aerosp. Sci. 2 (1), 1–95. https://doi.org/10.1016/0376-0421(62)90014-3.
Spalart, P., Allmaras, S., 1992. A one-equation turbulence model for aerodynamic flows. 30th Aerospace Sciences Meeting and Exhibit. American Institute of Aeronautics and Astronauticshttps://doi.org/10.2514/6.1992-439. Aerospace Sciences Meetings
Spalding, D.B., 1961. A single formula for the law of the wall. J. Appl. Mech. 28 (3), 455–458. https://doi.org/10.1115/1.3641728.
Suga, K., Craft, T., Iacovides, H., 2006. An analytical wall-function for turbulent flows and heat transfer over rough walls. International Journal of Heat and Fluid Flow 27 (5), 852–866. doi: j.ijheatfluidflow.2006.03.011. Special issue of the 6th International Symposium on Engineering Turbulence Modelling and Measurements ETMM6.